MSG No. 112.260 $P\bar{4}2c1'$ [Type II, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 222.1'

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	[1,2,3,4,9,10,11,12]
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	[5,6,7,8,13,14,15,16]

Table 2: Wyckoff site: 2b, site symmetry: 222.1'

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	[1,2,3,4,9,10,11,12]
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	[5,6,7,8,13,14,15,16]

Table 3: Wyckoff site: 2c, site symmetry: 222.1'

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[1,2,3,4,9,10,11,12]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[5,6,7,8,13,14,15,16]

Table 4: Wyckoff site: 2d, site symmetry: 222.1'

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	[1,2,3,4,9,10,11,12]
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	[5,6,7,8,13,14,15,16]

Table 5: Wyckoff site: 2e, site symmetry: -4..1'

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,4,5,6,9,12,13,14]
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16]

Table 6: Wyckoff site: 2f, site symmetry: -4..1'

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[1,4,5,6,9,12,13,14]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16]

Table 7: Wyckoff site: 4g, site symmetry: .2.1'

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,2,9,10]
2	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[3,4,11,12]

continued ...

Table 7

No.	position	mapping
3	$\left[0,\, -x,\, \frac{3}{4}\right]$	`[5,7,13,15]`
4	$\left[0,\, x,\, \frac{3}{4}\right]$	`[6,8,14,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[\frac{1}{2},\, y,\, \frac{1}{4}\right]$	`[1,3,9,11]`
2	$\left[\frac{1}{2},\, -y,\, \frac{1}{4}\right]$	`[2,4,10,12]`
3	$\left[y,\, \frac{1}{2},\, \frac{3}{4}\right]$	`[5,8,13,16]`
4	$\left[-y,\, \frac{1}{2},\, \frac{3}{4}\right]$	`[6,7,14,15]`

Table 9: Wyckoff site: `4i`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, \frac{1}{4}\right]$	`[1,2,9,10]`
2	$\left[-x,\, \frac{1}{2},\, \frac{1}{4}\right]$	`[3,4,11,12]`
3	$\left[\frac{1}{2},\, -x,\, \frac{3}{4}\right]$	`[5,7,13,15]`
4	$\left[\frac{1}{2},\, x,\, \frac{3}{4}\right]$	`[6,8,14,16]`

Table 10: Wyckoff site: `4j`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[0,\, y,\, \frac{1}{4}\right]$	`[1,3,9,11]`
2	$\left[0,\, -y,\, \frac{1}{4}\right]$	`[2,4,10,12]`
3	$\left[y,\, 0,\, \frac{3}{4}\right]$	`[5,8,13,16]`
4	$\left[-y,\, 0,\, \frac{3}{4}\right]$	`[6,7,14,15]`

Table 11: Wyckoff site: `4k`, site symmetry: `2..1'`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,4,9,12]`
2	$\left[0,\, 0,\, \frac{1}{2} - z\right]$	`[2,3,10,11]`
3	$[0,\, 0,\, -z]$	`[5,6,13,14]`
4	$\left[0,\, 0,\, z + \frac{1}{2}\right]$	`[7,8,15,16]`

Table 12: Wyckoff site: `4l`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[\frac{1}{2},\, \frac{1}{2},\, z\right]$	`[1,4,9,12]`
2	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2} - z\right]$	`[2,3,10,11]`
3	$\left[\frac{1}{2},\, \frac{1}{2},\, -z\right]$	`[5,6,13,14]`

continued ...

Table 12

No.	position	mapping
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8,15,16]`

Table 13: Wyckoff site: `4m`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,9,12]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[2,3,10,11]`
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[5,6,13,14]`
4	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[7,8,15,16]`

Table 14: Wyckoff site: `8n`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,9]`
2	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	`[2,10]`
3	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[3,11]`
4	$[-x,\ -y,\ z]$	`[4,12]`
5	$[y,\ -x,\ -z]$	`[5,13]`
6	$[-y,\ x,\ -z]$	`[6,14]`
7	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	`[7,15]`
8	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[8,16]`